

# **REPORT PAPER**

## **Customer Insights & KPI Dashboard**

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### **1. Overview :**

This project focuses on analyzing e-commerce sales data to identify key business insights, track performance metrics, and support data-driven decision-making. The dashboard was developed using Microsoft Power BI to visualize KPIs, customer behavior, and sales trends.

### **2. Objective :**

- To analyze overall sales performance.
- To identify top-performing products and customers.
- To understand customer demographics and behavior.
- To track monthly growth and trends.
- To derive actionable business insights.

### **3. Dataset Description :**

The dataset includes the following key fields:

- Customer ID
- Gender
- Age
- Product Name
- Category
- Unit Price
- Quantity
- Total Price
- Shipping Status
- Region
- Order Date

## 4. Data Cleaning & Preparation :

The data was processed using Python (Pandas) and prepared for analysis:

- Removed missing/null values
- Converted data types (Date, Numeric fields)
- Created new calculated fields:
  - Age Group
  - KPIs (Revenue, Customers, Orders)
- Ensured consistency and accuracy

## 5. Key Performance Indicators (KPIs) :

The following KPIs were created:

- **Total Revenue** – Overall sales generated
- **Total Orders** – Number of transactions

- **Total Customers** – Unique customers
- **Average Order Value (AOV)** – Avg. spend per order
- **Total Quantity Sold**

## 6. Dashboard Overview :

### Page 1: KPI & Sales Overview

- KPI Cards with trend (mini line charts)
- Monthly Revenue Trend
- Revenue vs Orders (Combo Chart)
- Sales by Region & Category
- Interactive filters

### Page 2: Customer & Product Insights

- Top Products Analysis
- Sales by Gender
- Sales by Age Group
- Shipping Status Distribution
- Top Customers Analysis
- Key Influencers (AI-based insights)
- Gauge Chart (Top customer contribution)

## 7. Key Insights :

- Electronics category generates the highest revenue.
- The 26–35 age group contributes the most to sales.
- The West region shows strong performance.
- A small number of customers contribute a large portion of revenue.

- Most orders are successfully delivered.
- Sales show consistent monthly growth trends.

## 8. Business Recommendations :

- Focus marketing efforts on high-performing categories.
- Target the 26–35 age segment for better conversion.
- Strengthen operations in high-performing regions.
- Improve retention strategies for top customers.
- Optimize delivery process to reduce delays.

## 9. Tools & Technologies Used :

Python (Pandas, NumPy)

**Microsoft Power BI**

Excel

GitHub (for project hosting)

## 10. Conclusion :

This project demonstrates the ability to transform raw data into meaningful insights using data analysis and visualization techniques. The dashboard enables stakeholders to monitor performance, understand customer behavior, and make informed business decisions.